Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two particles of masses m and $4m$ have linear momenta in the ratio of $2:1$. What is the ratio of their kinetic energies?

(A) $\sqrt{2}$

(B) 2

(C) 4

(D) 16

Q.2

A particle of mass m has half the kinetic energy of another particle of mass $m/2$. If the speed of the heavier particle is increased by $2\ \mathrm{ms}^{-1}$, its new kinetic energy equals the original kinetic energy of the lighter particle. The ratio of the original speeds of the lighter and heavier particles is

(A) $1:1$

(B) $1:2$

(C) $1:3$

(D) $1:4$

Q.3

A body of mass m accelerates uniformly from rest to velocity v_1 in time interval T_1. The instantaneous power delivered to the body as a function of time t is:

(A) $\frac{\mathrm{mv_1^2}}{\mathrm{T_1^2}}\mathrm{t}$

(B) $\frac{\mathrm{mv_1}}{\mathrm{T_1^2}}\mathrm{t}$

(C) $\left(\frac{\mathrm{mv_1}}{\mathrm{T_1}}\right)^2\mathrm{t}$

(D) $\frac{\mathrm{mv_1^2}}{\mathrm{T_1}}\mathrm{t}^2$

Q.4

The distance x moved by a body of mass $0.5\ \mathrm{kg}$ by a force varies with time t as
$x = 3t^2 + 4t + 5$
where x is expressed in metre and t in second. What is the work done by the force in the first 2 seconds?

(A) $25\ \mathrm{J}$

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(B) 50 J

(C) 60 J

(D) 100 J

Q.5

A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time t is proportional to

(A) $t^{3/4}$

(B) $t^{3/2}$

(C) $t^{1/4}$

(D) $t^{1/2}$

Q.6

A small coin is placed at a distance r from the centre of a gramophone record. The rotational speed of the record is gradually increased. If the coefficient of friction between the coin and the record is μ, the minimum angular frequency of the record for which the coin will fly off is given by

(A) $\sqrt{\frac{2\mu g}{r}}$

(B) $\sqrt{\frac{\mu g}{2r}}$

(C) $\sqrt{\frac{\mu g}{r}}$

(D) $2\sqrt{\frac{\mu g}{r}}$

Q.7

A pendulum has a length l. Its bob is pulled aside from its equilibrium position through an angle α and then released. The speed of the bob when it passes through the equilibrium position is given by

(A) $\sqrt{2gl}$

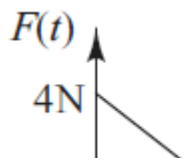
(B) $\sqrt{2gl\cos\alpha}$

(C) $\sqrt{2gl(1-\cos\alpha)}$

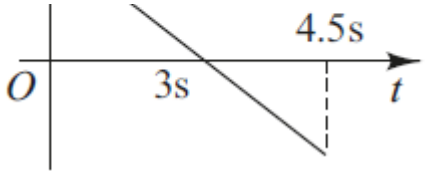
(D) $\sqrt{2gl(1-\sin\alpha)}$

Q.8

A block of mass 2 kg is free to move along the x-axis. It is at rest and from $t=0$ onwards it is subjected to a time-dependent force $F(t)$ in the x direction. The force $F(t)$ varies with t as shown in the figure. The kinetic energy of the block after 4.5 seconds is



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(A) 4.50 J

(B) 7.50 J

(C) 5.06 J

(D) 14.06 J

Q.9

A spring of spring constant $5 \times 10^3 \mathrm{Nm}^{-1}$ is stretched initially by 5 cm from the unstretched position. Then the work required to stretch by another 5 cm is

(A) 12.50 J

(B) 18.75 J

(C) 25.00 J

(D) 6.25 J

Q.10

A body of mass m is dropped from a certain height. It has a velocity v when it is at a height h above the ground. Which of the following will remain constant during the free fall?

(A) $v^2 + 2gh$

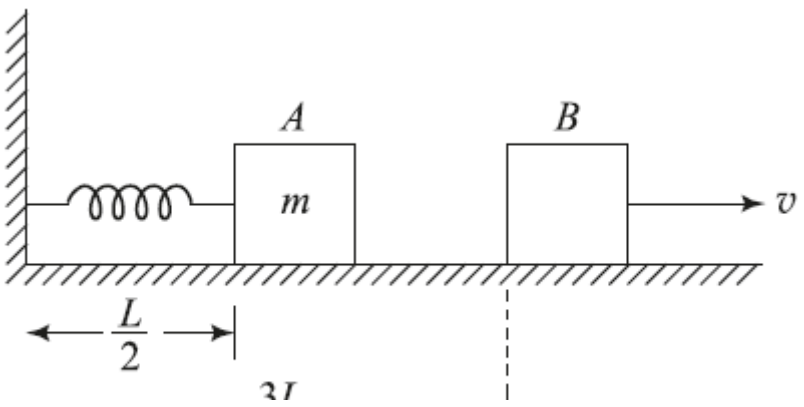
(B) $v^2 - 2gh$

(C) $v + \sqrt{\frac{gh}{2}}$

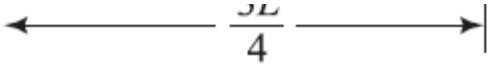
(D) $v - 2\sqrt{gh}$

Q.11

One end of an elastic spring of natural length L and spring constant k is fixed to a wall and the other end is attached to a block of mass m lying on a horizontal frictionless table (Fig. 4.9). The block is moved to a position A so that the spring is compressed to half its natural length and then released. What is the velocity of the block when it reaches position B which is at a distance $\frac{3L}{4}$ from the wall.



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(A) $\frac{L}{4}\sqrt{\frac{k}{m}}$

(B) $\frac{L}{2}\sqrt{\frac{2k}{m}}$

(C) $\frac{L}{4}\sqrt{\frac{3k}{m}}$

(D) $\frac{L}{2}\sqrt{\frac{k}{m}}$

Q.12

A bullet is fired from a rifle which recoils after firing. The ratio of the kinetic energy of the rifle to that of the bullet is

(A) zero

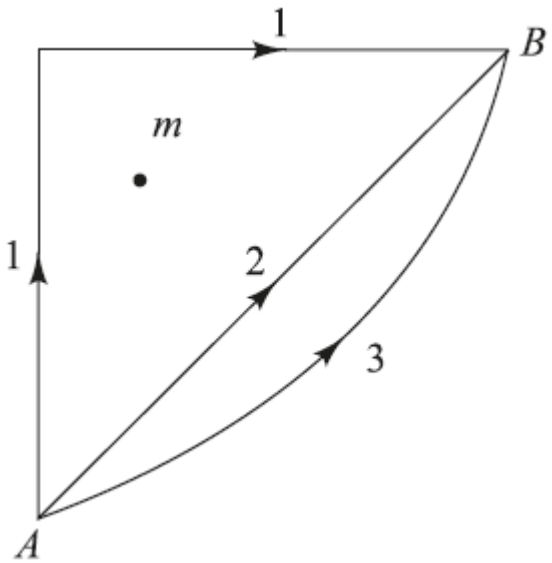
(B) one

(C) less than one

(D) more than one

Q.13

If W_1, W_2 and W_3 represent the work done in moving a particle from A to B along three different paths 1 , 2 and 3 (as shown in the figure) in the gravitational field of a point mass m, find the correct relation between W_1, W_2 and W_3



(A) $W_1 > W_3 > W_2$

(B) $W_1 = W_2 = W_3$

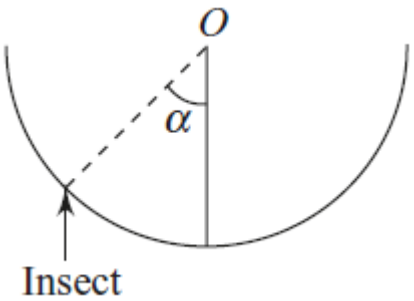
(C) $W_1 < W_3 < W_2$

(D) $W_1 < W_2 < W_3$

Q.14

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An insect crawls up a hemispherical surface very slowly (see Fig.). The coefficient of friction between the insect and the surface is $1/3$. If the line joining the center of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α is given by



(A) $\cot\alpha = 3$

(B) $\tan\alpha = 3$

(C) $\sec\alpha = 3$

(D) $\text{cosec}\,\alpha = 3$

Q.15

A force applied by an engine of a train of mass $2.05 \times 10^6\ \text{kg}$ changes its velocity from $5\ \text{m/s}$ to $25\ \text{m/s}$ in 5 minutes. The power of the engine is

(A) 1.025MW

(B) 2.05MW

(C) 5MW

(D) 6MW

Q.16

A body of mass $M\ \text{kg}$ is on the top point of a smooth hemisphere of radius $5\ \text{m}$. It is released to slide down the surface of the hemisphere. It leaves the surface when its velocity is $5\ \text{m/s}$. At this instant the angle made by the radius vector of the body with the vertical is: (Acceleration due to gravity $= 10\ \text{ms}^{-2}$)

(A) 30°

(B) 45°

(C) 60°

(D) 90°

Q.17

A body of mass $0.5\ \text{kg}$ is whirled in a vertical circle at an angular frequency of 10rads^{-1}. If the radius of the circle is $0.5\ \text{m}$, what is the tension in the string when the body is at the top of the circle? Take $g = 10\ \text{ms}^{-2}$

(A) $10\ \text{N}$

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(B) $20\ \mathrm{N}$

(C) $30\ \mathrm{N}$

(D) $40\ \mathrm{N}$

Q.18

The pilot of an aircraft, who is not tied to his seat, can loop a vertical circle in air without falling out at the top of the loop. What is the minimum speed required so that he can successfully negotiate a loop of radius $4\ \mathrm{km}$? Take $g = 10\ \mathrm{ms}^{-2}$

(A) $100\ \mathrm{ms}^{-1}$

(B) $200\ \mathrm{ms}^{-1}$

(C) $300\ \mathrm{ms}^{-1}$

(D) $400\ \mathrm{ms}^{-1}$

Q.19

A car of mass ' m ' is driven with acceleration ' a ' along a straight level road against a constant external resistive force ' R '. When the velocity of the car is ' V, the rate at which the engine of the car is doing work will be

(A) RV

(B) maV

(C) $(R + ma)V$

(D) $(ma - R)V$

Q.20

An engine develops $10\ \mathrm{kW}$ of power. How much time will it take to lift a mass of $200\ \mathrm{kg}$ to a height of $40\ \mathrm{m} (g = 10\ \mathrm{m/sec}^2)$?

(A) 4sec

(B) 5sec

(C) 8sec

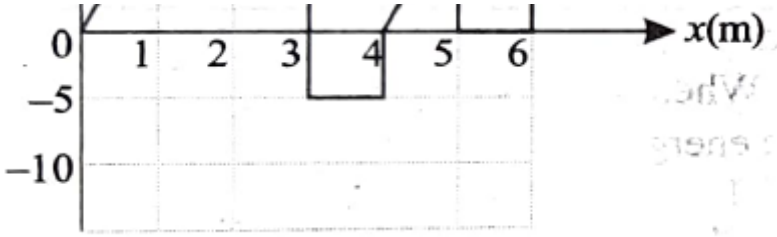
(D) 10sec

Q.21

The relationship between the force F and position x of a body is as shown in figure. The work done in displacing the body from $x = 1\ \mathrm{m}$ to $x = 5\ \mathrm{m}$ will be



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Q.22

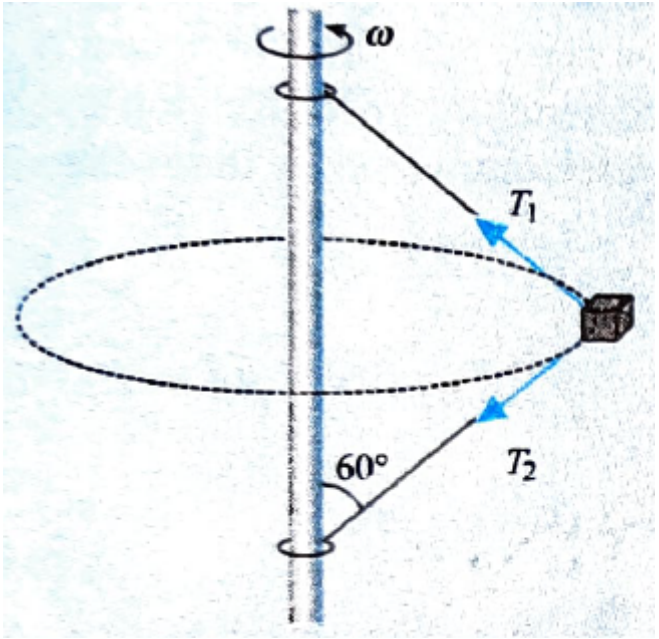
In a shotput event an athlete throws the shotput of mass 10 kg with an initial speed of $1\ \mathrm{m\ s^{-1}}$ at 45° from a height $1.5\ \mathrm{m}$ above ground. Assuming air resistance to be negligible and acceleration due to gravity to be $10\ \mathrm{m\ s^{-2}}$, the kinetic energy of the shotput when it just reaches the ground will be

Q.23

A constant force $\mathbf{F} = (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}})$ newton displaces a body from position $\mathbf{r}_1 = (4\hat{\mathbf{i}} - 5\hat{\mathbf{j}})$ metre to $\mathbf{r}_2 = (\hat{\mathbf{i}} + 3\hat{\mathbf{j}})$ metre. Find the work done by the force.

Q.24

A small block is connected to one end of two identical passless strings of length $16\frac{2}{3}\ \mathrm{cm}$ each with their other ends fred to a vertical rod. If the ratio of tensions T_1/T_2 be $4:1$, ten what will be the angular velocity ω of the block?



Q.25

A car of mass $500\ \mathrm{kg}$ moving with a speed $36\mathrm{kmh}$ in a straight road unidirectionally doubles its speed in $1\ \mathrm{min}$. Find the power delivered by the engine.

Q.26

A block of mass $5\ \mathrm{kg}$ is being raised vertically upwards by the help of a string attached to it. It rises with an acceleration of $2\ \mathrm{m\ s^{-2}}$. The tension in the string if the block rises by $2.5\ \mathrm{m}$. Find the net work done.

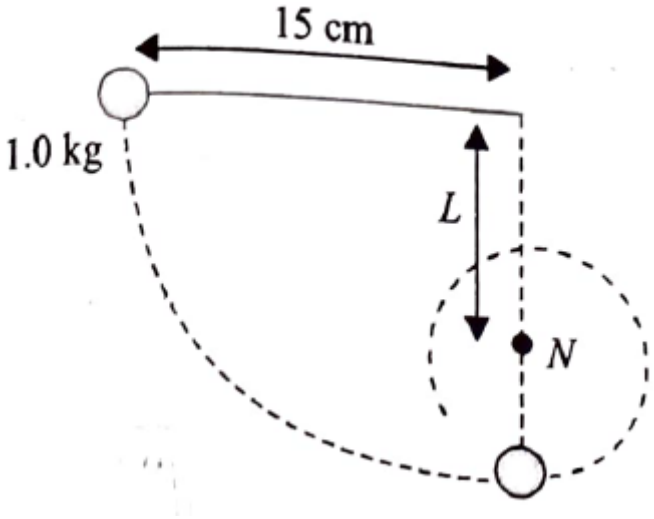
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Q.27

A bomb of mass $16\ \text{kg}$ at rest explodes into two pieces of masses $4\ \text{kg}$ and $12\ \text{kg}$. The velocity of the $12\ \text{kg}$ mass is $4\ \text{ms}^{-1}$. The kinetic energy of the other mass is

Q.28

A ball weighing $1.0\ \text{kg}$ is tied to a string $15\ \text{cm}$ long. initially the ball is held in position such that the (initially) the ball is held in position such that the string is horizontal. The ball is now released. A nail N is situated vertically below the support at a distance L. The minimum value of L such that the string will be wound round the nail is



Q.29

A block of mass $m = 2\ \text{kg}$ is raised vertically upwards by means of a massless string through a distance of $S = 4\ \text{m}$ with a constant acceleration $a = 2.2\ \text{ms}^{-2}$. Find the work done by tension.

Q.30

A stone of mass $1\ \text{kg}$ tied to a light inextensible string of length $L = \frac{10}{3}\ \text{m}$ is whirling in a circular path of radius L in a vertical plane. If the ratio of the maximum tension in the string to the minimum tension in the string is 4 and if g is taken to be $10\ \text{m/sec}^2$, the speed of the stone at the highest point of the circle is

Answers & Solutions

Q.1 Answer:

16

Solution:

Given $p_1 = m_1v_1 = 2p$ and $p_2 = m_2v_2 = p$, so that
$\frac{m_1v_1}{m_2v_2} = 2$
The ratio of their kinetic energies is
$\frac{(\text{KE})_1}{(\text{KE})_2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_2v_2^2} = \frac{m_1^2v_1^2}{m_2^2v_2^2} \cdot \frac{m_2}{m_1}$
But $m_2 = 4m_1$ and $\frac{m_1v_1}{m_2v_2} = 2$. Therefore,

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$\frac{(KE)_1}{(KE)_2} = (2)^2 \times 4 = 16$

Hence the correct choice is (d).

Q.2 Answer:

$1 : 2$

Solution:

$$\frac{1}{2}mv_1^2 = \frac{1}{2}\left(\frac{1}{2}\left(\frac{m}{2}\right)v_2^2\right)$$

$$v_2 = 2v_1$$

$$v_1 = 2(1+\sqrt{2})$$

$$v_2 = 4(1+\sqrt{2})$$

$$k\varepsilon_1 : k\varepsilon_2 = 1 : 2$$

Q.3 Answer:

$\frac{mv_1^2}{T_1^2}t$

Solution:

Let a be the constant acceleration

Using 1st equation of motion

$$v_1 = u + aT_1$$

$$v_1 = 0 + aT_1$$

$$a = \frac{v_1}{T_1} \ldots\ldots\ldots \text{(i)}$$

At any instant t, the velocity v of the body

$$v = 0 + at$$

$$v = \frac{v_1}{T_1} \cdot t \ldots\ldots\ldots \text{(ii)}$$

$$\because P = Fv$$

$$P = mav$$

$$P = m\left(\frac{v_1}{T_1}\right)\left(\frac{v_1}{T_1} \cdot t\right)$$

$$P = \frac{mv_1^2}{T_1^2}t$$

Q.4 Answer:

60 J

Solution:

Velocity $(v) = \frac{dx}{dt} = \frac{d}{dt}\left(3t^2 + 4t + 5\right) = 6t + 4$

Acceleration is $a = \frac{dv}{dt} = \frac{d}{dt}(6t + 4) = 6\ \text{ms}^{-2}$

Therefore, applied force is $F = ma = 0.5 \times 6 = 3$ N.

Now $t = 2$ s**, the distance moved is** $x = 3 \times (2)^2 + 4 \times 2 + 5 = 25$ m $\quad (\because u = 4\ \text{ms}^{-1}$ **and**

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$x = 5$ m at $t = 0$)

Hence the correct choice is (c).

Q.5 Answer:

$t^{3/2}$

Solution:

$$P = Fv = mav = m\frac{dv}{dt}v = m\frac{dv}{dx}\cdot\frac{dx}{dt}v = mv^2\frac{dv}{dx}$$
$$\therefore \quad v^2dv = \frac{P}{m}dx$$

Integrating (assuming that the body starts from rest), we have

$$\int_0^v v^2dv = \frac{P}{m}\int_0^x dx \quad (\because P = \text{constant})$$
$$\Rightarrow \quad \frac{v^3}{3} = \frac{Px}{m}$$
$$\Rightarrow \quad \frac{v^3}{3} = \frac{Px}{m}$$
$$\Rightarrow v = \left(\frac{3P}{m}\right)^{1/3} = x^{1/3} = kx^{1/3} \text{ where } k = \left(\frac{3P}{m}\right)^{1/3} = \textbf{constant}$$
$$\Rightarrow \quad \frac{dx}{dt} = kx^{1/3}$$
$$\Rightarrow \quad x^{-1/3}dx = kdt$$

Integrating $\int_0^x x^{-1/3}dx = k\int_0^t dt$

$$\Rightarrow \quad \frac{x^{2/3}}{2/3} = kt \Rightarrow x = \left(\frac{2k}{3}\right)^{3/2} \times t^{3/2}$$

Hence $\quad x \propto t^{3/2}$

Q.6 Answer:

$\sqrt{\frac{\mu g}{r}}$

Solution:

Let m be the mass of the coin. It will fly off when the centripetal force $mr\omega^2$ just exceeds the force of friction μmg. The minimum ω is given by

$$mr\omega^2 = \mu mg$$
$$\omega = \sqrt{\frac{\mu g}{r}}$$

Hence the correct choice is (c).

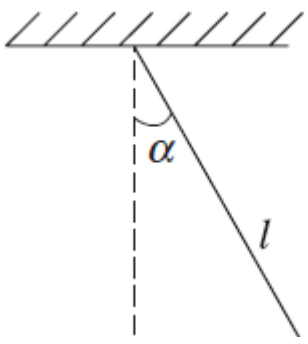
Q.7 Answer:

$\sqrt{2gl(1-\cos\alpha)}$

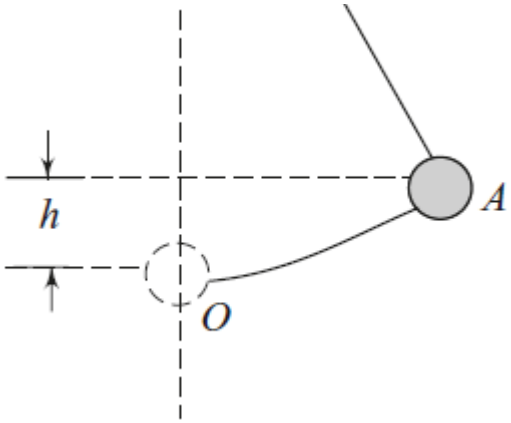
Solution:

As shown in Fig., the height attained by the bob when the string subtends an angle α with the vertical is

$$h = l - l\cos\alpha = l(1-\cos\alpha)$$



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Its potential energy at the highest point $A = mgh$ where m is the mass of the bob. Let v be the speed of the bob when it passes through O. Its kinetic energy at $O = \frac{1}{2}mv^2$. From the principle of conservation of energy, we have

$\frac{1}{2}mv^2 = mgh$

or $v = \sqrt{2gh} = \sqrt{2gl(1 - \cos\alpha)}$

Hence the correct choice is (c).

Q.8 Answer:

5.06 J

Solution:

Slope of graph $= -\frac{4\text{ N}}{3\text{ s}} = -\frac{4}{3}\text{Ns}^{-1}$. **Therefore,**

$F = -\frac{4}{3}t + 4$. Now change in momentum $= \int F dt$

$\Rightarrow \quad mv - 0 = \int_0^{4.55} \left(-\frac{4}{3}t + 4\right) dt$

or $2v = \left|-\frac{2t^2}{3} + 4t\right|_0^{4.5s} = 4.5$

Which gives $v = 2.25\text{ ms}^{-1}$. **Therefore,**

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times (2.25)^2$$
$$\simeq 5.06\text{ J, which is choice (c)}$$

Q.9 Answer:

18.75 J

Solution:

Here $x_1 = 5\text{ cm} = 0.05\text{ m}$ **and** $x_2 = 10\text{ cm} = 0.1\text{ m}$

$$W = \frac{1}{2}k\left(x_2^2 - x_1^2\right)$$
$$= \frac{1}{2} \times 5 \times 10^3 \left[(0.1)^2 - (0.5)^2\right] = 18.75\text{ J}$$

Q.10 Answer:

$v^2 + 2gh$

Solution:

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The total energy $= \mathrm{KE} + \mathrm{PE}$ remains constant during the free fall, i.e.

$mgh + \frac{1}{2}mv^2 = \text{constant}$

or $gh + \frac{v^2}{2} = \text{constant}$

Hence the correct choice is (a).

Q.11 Answer:

$\frac{L}{4}\sqrt{\frac{3k}{m}}$

Solution:

At position A, the compression of the spring is $\frac{L}{2}$. At position B, the compression is $L - \frac{3L}{4} = \frac{L}{4}$. Therefore, loss of potential energy as the block moves from A to B is given by

$\frac{1}{2}k\left(\frac{L}{2}\right)^2 - \frac{1}{2}k\left(\frac{L}{4}\right)^2 = \frac{3kL^2}{32}$

From conservation of energy, loss in P.E. = gain in K.E., i.e.

$$\frac{3kL^2}{32} = \frac{1}{2}mv^2$$

$$\Rightarrow v = \frac{L}{4}\sqrt{\frac{3k}{m}}$$

Q.12 Answer:

less than one

Solution:

Let M be the mass of the rifle and m that of the bullet and let V and v be their respective speeds. From the principle of conservation of momentum, we have $MV = mv$. The ratio of their kinetic energies is

$\frac{K_r}{K_b} = \frac{\frac{1}{2}MV^2}{\frac{1}{2}mv^2} = \frac{MV^2}{mv^2} = \frac{(MV)^2}{(mv)^2} \cdot \frac{m}{M} = \frac{m}{M}$

Since $m < M, K_r < K_b$. Hence the correct choice is (c).

Q.13 Answer:

$W_1 = W_2 = W_3$

Solution:

Gravitational force is conservative. The work done by a conservative force on a particle moving between two points does not depend on the path taken by the particle. Hence the correct choice is (b).

Q.14 Answer:

$\cot\alpha = 3$

Solution:

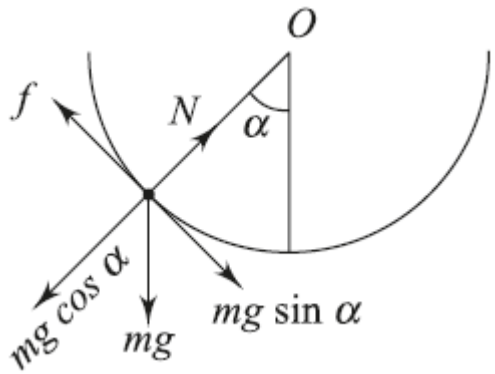
As shown in Fig., the insect will crawl without slipping if the value of α is not greater than that given by the condition: force of friction $f = mg\sin\alpha$. Now $f = \mu N$, where N is the normal reaction. Thus

$\mu N = mg\sin\alpha$

or $\mu mg\cos\alpha = mg\sin\alpha$

or $\cot\alpha = \frac{1}{\mu} = 3$, **which is choice (a).**

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Q.15 Answer:

$2.05\mathrm{MW}$

Solution:

Given that,

Mass of train, $m = 2.05 \times 10^6\ \mathrm{K\ g}$

Time, $t = 5$ **min utes** $= 300\ \mathrm{s}$

$\mathrm{v} = 25\ \mathrm{m/s}$

$\mathrm{u} = 5\ \mathrm{m/s}$

Acceleration,

$\mathrm{a} = \frac{v-\mathrm{u}}{\mathrm{t}}$

$\mathrm{a} = \frac{25-5}{300}$

$\mathrm{a} = \frac{2}{30} = \frac{1}{15}\ \mathrm{m/s^2}$

Using equation of motion,

$\gamma^2 - u^2 = 2\ \mathrm{ns}$

$(25)^2 - (5)^2 = 2 \times \frac{1}{15} \times \mathrm{s}$

$\mathrm{s} = 4500\ \mathrm{m}$

Power,

$\mathrm{P} = \frac{\text{work done}}{\text{time}}$

$\mathrm{P} = \frac{\mathrm{F}\times\mathrm{d}}{\mathrm{t}}$

$\mathrm{P} = \frac{\mathrm{m}\times\mathrm{a}\times\mathrm{s}}{\mathrm{t}}$

$\mathrm{P} = \frac{2.05\times10^6\times1\times4500}{15\times300}$

$\mathrm{P} = 2.05 \times 10^6\ \mathrm{W}$

$\mathrm{P} = 2.05\mathrm{MW}$

Q.16 Answer:

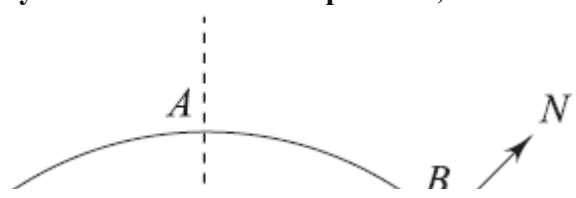
60°

Solution:

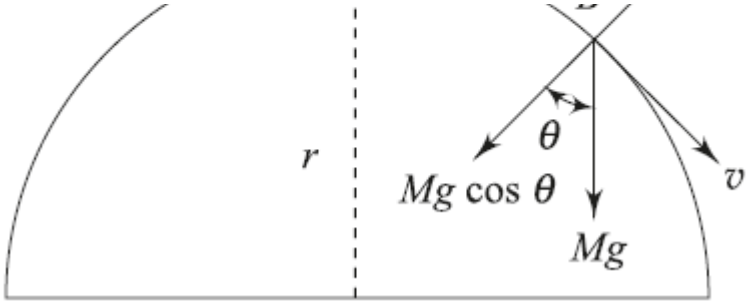
Let the body leave the surface at point B **as shown in Fig. When the body is between points** A **and** B**, we have**

$Mg\cos\theta - N = \frac{Mv^2}{r}$

When the body leaves the surface at point B**, the normal reaction** N **becomes zero. Thus**



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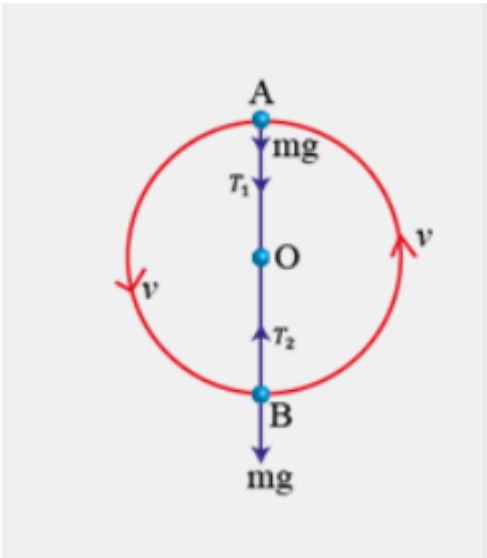
$Mg\cos\theta = \frac{Mv^2}{r}$

or $\cos\theta = \frac{v^2}{rg} = \frac{(5)^2}{5\times 10} = \frac{1}{2}$ **or** $\theta = 60^\circ$ **Hence the correct choice is (c).**

Q.17 Answer:

20 N

Solution:



It is clear from Fig., that when the body at the top point A of the circle, its weight mg and tension T_1 in the string act downwards towards the centre O of the circle and the sum of the two provides the necessary centripetal force. Thus

$$T_1 + mg = mR\omega^2$$

$$\text{Or } T_1 = m\left(R\omega^2 - g\right)$$

$$= 0.5 \times \left(0.5 \times 10^2 - 10\right)$$

$$= 20 \text{ N}$$

Thus, the correct choice is (b)

Q.18 Answer:

200 ms^{-1}

Solution:

The pilot does not drop down when he is at the top of the loop because his weight mg is less than the centripetal force mv^2/R required to keep him in the loop. The rest of the centripetal force is balanced by the reaction of the seat. Hence, he is stuck to the seat without being tied to it. If the speed of the aircraft is

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reduced so that $mg > mv^2/R$, he will fall off from his seat. Therefore, the minimum speed $v_{\min}$ required to successfully negotiate the vertical loop is given by

$$mg = \frac{mv_{\min}^2}{R}$$
$$\text{or } v_{\min} = \sqrt{gR} = \sqrt{10 \times 4000}$$
$$= 200 \text{ ms}^{-1}$$

Thus, the correct choice is (b).

Q.19 Answer:

$(R + ma)V$

Solution:

Total force require to oppose the resistive force and move with acceleration a is,

$\mathrm{F} = \mathrm{ma} + \mathrm{R}$

Power is given by,

$\therefore \mathrm{P} = (\mathrm{ma} + \mathrm{R})\mathrm{V}$

Q.20 Answer:

8sec

Solution:

$t = \frac{mgh}{p}$
$= \frac{200 \times 40 \times 10}{10 \times 10^3}$
$= 8$ secs

Q.21 Answer:

—

Solution:

Work done = area under $\mathrm{F} - \mathrm{x}$ graph = area of rectangle $\mathrm{ABDC}+$ area of rectangle $\mathrm{BCEF}+$ area of rectangle $\mathrm{GFIH}+$ area of triangle IJK

$$= (2-1) \times (10-0) + (3-2)(5-0)$$
$$+ (4-3)(-5-0) + \frac{1}{2}(5-4)(10-0)$$
$$= 10 + 1 \times 5 + (-5) + \frac{1}{2} \times 10 = 15J$$

Q.22 Answer:

—

Solution:

Initial KE of the shotput

$= \frac{1}{2}(10 \text{ kg})\left(1 \text{ m/s}^2\right) = 5 \text{ J}$

Initial PE of the shotput at a height 1.5 m abore ground $= \mathrm{mgh} =$

$(10 \text{ kg})\left(10 \text{ m/s}^2\right)(1.5 \text{ m}) = 150 \text{ J}$

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Total initial energy of the shotput $= 155$ J

Since air resistance is negligible and final potential of the shotput as it hits the ground $(\mathrm{h} = 0)$ **is zero, kinetic energy of the shotput on hitting the ground** $= 155$ J

Q.23 Answer:

—

Solution:

Displacement $\mathbf{S} = \mathbf{r}_2 - \mathbf{r}_1$

$= (\hat{\mathbf{i}} + 3\hat{\mathbf{j}}) - (4\hat{\mathbf{i}} - 5\hat{\mathbf{j}}) = -3\hat{\mathbf{i}} + 8\hat{\mathbf{j}}$

$\therefore \quad W = \mathbf{F} \cdot \mathbf{S} = (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}) \cdot (-3\hat{\mathbf{i}} + 8\hat{\mathbf{j}})$

$= -6 + 24 = 18$ J

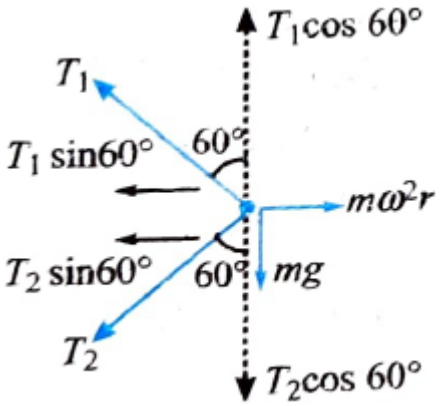
$[\because \hat{\mathbf{i}} \cdot \hat{\mathbf{i}} = \hat{\mathbf{j}} \cdot \hat{\mathbf{j}} = 1$ **and** $\hat{\mathbf{i}} \cdot \hat{\mathbf{j}} = 0]$

Q.24 Answer:

—

Solution:

For horizontal equilibrium of the block,



$T_1 + T_2 = m\omega^2 l$ **(i)**

For vertical equilibrium of the block,

$T_1 \cos 60° = T_2 \cos 60° + mg$ **(ii)**

Dividing Eqs. (i) by (ii),

$\frac{T_1+T_2}{T_1-T_2} = \frac{\omega^2 l}{2g} \Rightarrow \frac{\frac{T_1}{T_2}+1}{\frac{T_1}{T_2}-1} = \frac{\omega^2 l}{2g}$

$\frac{4+1}{4-1} = \frac{\omega^2 l}{2g}$

$\omega^2 = \frac{10g}{3l} = \frac{10 \times 9.8 \times 3}{3 \times 50 \times 10^{-2}} = 196$

$\omega = 14 \mathrm{rads}^{-1}$

Q.25 Answer:

—

Solution:

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Its initial speed $v_1 = \frac{36000}{3600} = 10 \text{ m s}^{-1}$.

If the car doubles its speed, finally its speed becomes $v_2 = 20 \text{ ms}^{-1}$.

change in KE of the car $= \Delta \text{KE} = (1/2)mv_2^2 - (1/2)mv_1^2$

the power delivered by the engine,

$$p = \frac{\text{Work done}}{\text{Time taken}} = \frac{|\Delta KE|}{\Delta t}$$
$$= \frac{\frac{1}{2} \times m\left[v_2^2 - v_1^2\right]}{\Delta t}$$
$$= \frac{1/2 \times 500\left[20^2 - 10^2\right]}{60} = 1250 \text{ W}$$

Q.26 Answer:

—

Solution:

Let us first calculate the tension. From figure $T - mg = 5a; T = 5(10 + 2) = 60 \text{ N}$

As T and displacement are in same direction (upwards), work done by the tension T is W.

$W = Ts = 60(2.5) = 150 \text{ J}$

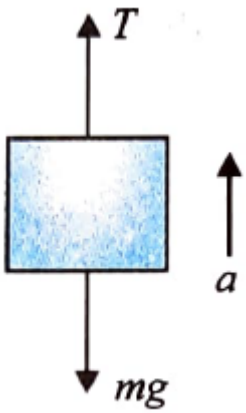
Work done by gravity

$= -mgs = -5(10)(2.5) = -125 \text{ J}$

Net work done on the block

$= \text{ Work done by } T + \text{ Work done by } mg$

$= 150 + (-125) = 25 \text{ J}$



Q.27 Answer:

—

Solution:

Let v be the speed of the 4 kg mass just after the explosion. Since the bomb was at rest, its momentum is zero. From the conservation of momentum, we have

$4 \times v - 12 \times 4 = 0 \Rightarrow v = 12 \text{ ms}^{-1}$

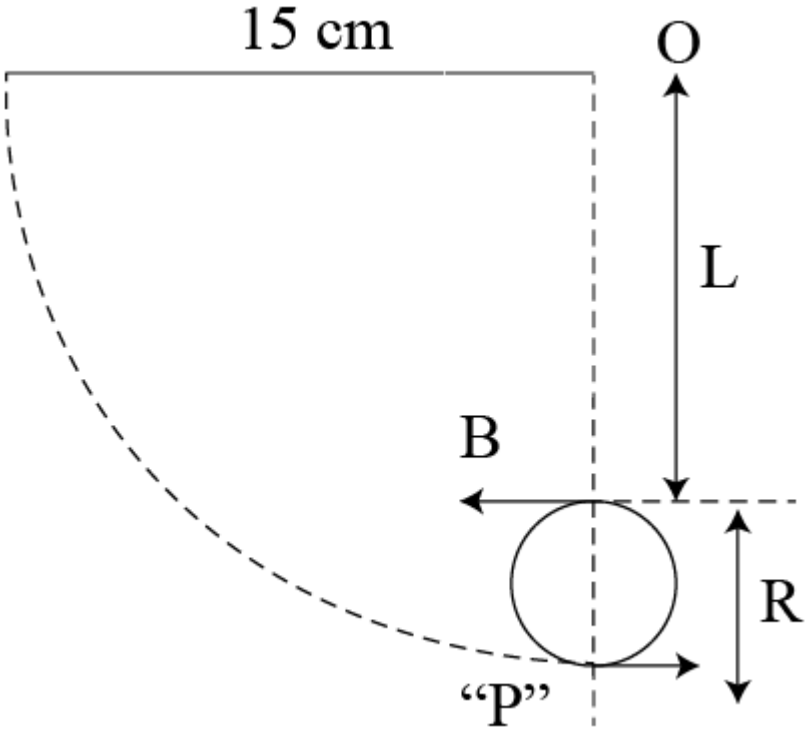
$\therefore$ K.E. of 4 kg mass $= \frac{1}{2} \times 4 \times (12)^2 = 288 \text{ J}$

Q.28 Answer:

—

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Solution:



Velocity at lowest point is $v\sqrt{5gR}$

Here at point $P, v = \sqrt{5g(R-L)}$ **(1)**

now, from Energy Conserivation law

$mgh = \frac{1}{2}mv^2$

$s_0, v^2 = 2gh$

Here, $h = R$

So, $v^2 = 2gR$

$v = \sqrt{2gR} - (2)$ **(2)**

From (1) & (2) we get

$\sqrt{5g(R-L)} = \sqrt{2gR}$

$\Rightarrow 5(R-L) = 2R$

$\Rightarrow 3R = 5L$

$L = \frac{3}{5}R = \frac{3}{5} \times 15 = 9 \text{ cm}$

Q.29 Answer:

—

Solution:

From the free body diagram

$T - mg = ma$

$\Rightarrow T = m(a+g)$

$= 2 \times (2.2 + 9.8)$

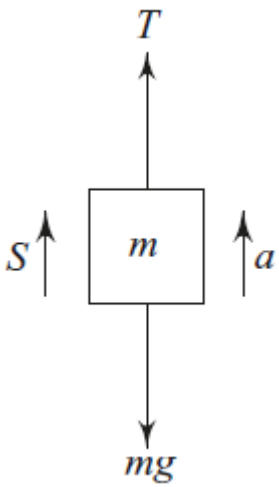
$= 24 \text{ N}$

∴ Work done by tension is $(\because T$ **and** S **are in the same direction)**

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$W_1 = TS\cos 0^\circ$

$= 24 \times 4 \times 1 = 96$ J



Q.30 Answer:

—

Solution:

Given that,

Mass m $= 1$ kg

Length $\mathrm{L} = \frac{10}{3}$ m

Radius $\mathrm{r} = 1$ m

Ratio of tension $\frac{T_2}{T_1} = 4$

Now,

The ratio of maximum T_2 **to the minimum tension is**

$\frac{\mathrm{T_2}}{\mathrm{T_1}} = 4$

$\mathrm{T_q} = 4\ \mathrm{T_1}$

Now, the difference between the two tensions should be $6\mathrm{mg}$ **Now,**

$\mathrm{T_2} - \mathrm{T_1} = 6\mathrm{ang}$

$4\ \mathrm{T_1} - \mathrm{T_1} = 6\mathrm{mg}$

$3\ \mathrm{T_1} = 6\mathrm{cmg}$

$\mathrm{T_1} = 2\mathrm{mg}$

Now, the tension at the top of the circle is

$\mathrm{T_1} + \mathrm{mg} = \frac{\mathrm{mv_1^2}}{\mathrm{r}}$

$T_1 + mg = \frac{mv^2}{L}$

$3\mathrm{lmg} = \frac{3\mathrm{mv_H^2}}{10}$

$v_1^2 = \frac{30\mathrm{mg}}{3\ \mathrm{m}}$

$\mathrm{v_1} = \sqrt{10 \times 10}$

$v_1 = 10\ \mathrm{m/s}$

Hence, the speed of stone at highest point of circle is $10\ \mathrm{m/s}$

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